Object-Grounded Text Priors for Saliency Ranking Detection

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1. Supplemental Materials

The supplement details the evaluation protocol and fairness considerations, including an InstructBLIP- and backbone-matched comparison with LG-SOR in Section 1.1. Section 1.2 presents additional caption analyses covering LVLM selection (Subsection 1.2.1), linguistic cues (Subsection 1.2.2), and description hallucinations (Subsection 1.2.3). We also report a manual audit of CLIP-based ROI-text matching in Section 1.3 and ablations of the MKTA kernel-gating factor in Section 1.4. Finally, Section 1.5 discusses representative failure cases, while Section 1.6 provides additional qualitative comparisons. Our code is publicly available at https://anonymous.4open.science/r/MSR-4C25.

1.1. Evaluation Protocol and Fairness Considerations

1.1.1. Evaluation Protocol

We evaluate each baseline under its original inference setting as closely as possible. Each method is provided with the type of input it natively requires, such as image-only input or object proposals generated by its corresponding proposal mechanism [4, 15, 19, 21], following the protocol used in the original work. Our method is evaluated under the same principle. As shown in Table 1, the compared methods differ in both input requirements and output formats. To ensure fairness, we therefore do not force all methods into a unified candidate-instance protocol that deviates from their original design. Instead, each method is evaluated using its native inference pipeline, while all methods are compared using the same saliency-ranking metrics. This protocol reflects end-to-end performance under each method's native design, but it does not isolate the ranking module from differences in proposal or segmentation quality.

Most baselines are instance-level saliency ranking methods, except RSDNet. RSDNet directly predicts pixel-level saliency maps rather than explicit saliency rank scores or ranked instance masks. Following the official RSDNet/SOR protocol, object-level scores are obtained by averaging predicted saliency values within each ground-truth object mask, and objects are ranked by these averages. Since RSDNet does not predict instance masks, its official code does not compute SA-SOR directly; for SA-SOR evaluation, its saliency map must be paired with a specified instance-mask source. In our evaluation, we use the instance segmentation masks provided by the corresponding evaluation datasets for this purpose. Therefore, the SA-SOR result for RSDNet should be interpreted as evaluating the ranking induced by its saliency map over the provided instance masks, rather than end-to-end instance-mask prediction. By contrast, the remaining methods generate saliency maps from predicted ranks and segmentation masks. Among the saliency-ranking baselines, LG-SOR is the only multimodal method that uses image captions; all others are visual-only. For a fair comparison, we use LG-SOR's best-performing caption generator, InstructBLIP [5], and follow its original protocol by limiting generated descriptions to 256 characters in all LG-SOR experiments.

For each baseline, we use the authors' official checkpoint when one is available for the corresponding dataset and evaluation split. Otherwise, we train the model from scratch using the authors' released code and recommended settings. Specifically, we evaluate LG-SOR using its official checkpoints on ASSR and IRSR and train it from scratch on SIFR, for which no compatible checkpoint is available.

The compared methods also differ in how ROIs are obtained for extracting object-level features. ASSR trains the FPN-based segmentation branch [15] and the saliency ranking branch separately, and uses proposals generated by the trained segmentation branch during both training and inference. IRSR uses jointly trained RPN-based proposal modules [21]. SOR and QAGNet likewise couple saliency rank-

Table 1. Comparison of baseline methods in terms of native input requirements, including RGB images (I), captions (C), and object proposal sources (ROI), together with output formats, including saliency rank scores (SR), segmentation masks (Masks), and saliency maps (Sal-Map). The table also reports the method category: pixel-level saliency ranking (Pixel) or instance-based saliency ranking (Inst.).

Model	Category		Training Input			Inference Input			Output		
	Pixel	Inst.	I	C	ROI	I	C	ROI	SR	Masks	Sal-Map
RSDNet [11]	✓		✓		-	✓		-			✓
ASSR [22]		✓	✓		FPN	✓		FPN	✓	✓	✓
IRSR [18]		✓	✓		RPN	✓		RPN	✓	✓	✓
SOR [7]		✓	✓		MaskRCNN	✓		MaskRCNN	✓	✓	✓
OCOR [24]		✓	✓		Box Queries	✓		Box Queries	✓	✓	✓
QAGNet [6]		✓	✓		Mask2Former	✓		Mask2Former	✓	✓	✓
LG-SOR [16]		✓	✓	✓	Mask2Former	✓	✓	Mask2Former	✓	✓	✓
Ours		✓	✓	✓	Mask2Former	✓	✓	Mask2Former	✓	✓	✓

ing with jointly trained instance-generation modules based on Mask R-CNN [8] and Mask2Former [4], respectively. LG-SOR similarly uses ROIs and Segmentation masks by training its Mask2Former segmentation branch.

Our method follows a decoupled proposal-and-ranking paradigm, similar to ASSR. In the first stage, the Mask2Former-based segmentation branch is trained to generate object region masks and ROIs, while the saliency ranking branch, including the TCSE module and backbone, is frozen. In the second stage, the saliency ranking branch is trained using the object proposals generated by the pretrained segmentation branch, which remains frozen throughout this stage.

1.1.2. Additional Comparison with LG-SOR

Since LG-SOR uses InstructBLIP captions, we additionally evaluate TSPNet with an InstructBLIP-based caption generator and a ResNet-50 backbone to match the reported LG-SOR setting as closely as possible. As shown in Table 2, TSPNet obtains higher scores than LG-SOR under this caption-generator/backbone-matched comparison, with relative improvements of 8.68% in SA-SOR and 1.45% in SOR on the ASSR dataset, and 20.37% in SA-SOR and 3.96% in SOR on the IRSR dataset. These results reduce the likelihood that TSPNet's gains are explained solely by the choice of caption generator, although they do not control for all implementation differences between the two methods. Together with the ablation results, the improvements are consistent with the contribution of the proposed ROI-level text-conditioned saliency estimation, spatial prior injection, and text-adaptive feature refinement mechanisms.

Table 2. Comparison with LG-SOR under matched caption-generation settings.

	ASSR		IRSR	
Model	**SA-SOR↑**	**SOR↑**	**SA-SOR↑**	**SOR↑**
LG-SOR	0.7360	0.8875	0.5690	0.8150
Ours	0.7999	0.9004	0.6849	0.8473

1.2. Additional Ablations

1.2.1. LVLM Analysis

Table 3 compares different LVLM caption generators for TSPNet on IRSR by separately training and evaluating TSPNet using captions from different caption generators. For both offline and online settings, captions are generated using two NVIDIA GeForce RTX 4090 GPUs with batch size of 64, following the same prompt and 77-token truncation used in the main experiments. All LVLMs are run with the same decoding settings and inference precision. We report caption generation time per image t_d in seconds, cached-caption inference time t_{off}, and online inference time with caption generation t_{on} in minutes over the full test split. In the cached setting, captions are reused during training and evaluation, so t_{off} measures only TSPNet inference with precomputed descriptions. In the online setting, captions are generated on-the-fly in batches of 64 and cached before TSPNet inference is performed with a batch size of 4. ShareGPT4V-7B [2] achieves the best performance while requiring 0.30 seconds per image for caption generation. Gemma3-12B [23] obtains the second-best SOR score, and LLaVA-7B [17] matches it in SA-SOR but gives lower SOR. Although Qwen2-VL-7B [25] is the fastest generator and has the lowest online inference time, its ranking accuracy is lower than that of the top-performing models. These results suggest that caption quality affects the usefulness of text-conditioned saliency priors. Comparable results across separately trained caption sources demonstrate robustness

to caption-generator choice. Since our main setting uses cached offline captions, the LVLM choice primarily affects the one-time preprocessing cost and online inference time, while cached-caption inference remains unchanged across caption generators. The cached captions are lightweight text files and will be released with the prompt and decoding settings to support reproducibility.

Table 3. Effect of LVLM caption generators and caption generation latency.

LVLM	SA-SOR↑	SOR↑	t_d (s)	t_{off} (m)	t_{on} (m)
InternVL3.5-8B [3]	0.6832	0.8444	0.32	1.65	17.29
LLaVA-7B [17]	0.7112	0.8500	0.38	1.65	20.22
Qwen2-VL-7B [25]	0.6967	0.8536	0.23	1.65	12.89
InstructBLIP-7B [5]	0.6975	0.8536	0.33	1.65	17.78
Gemma3-12B [23]	0.7112	0.8580	0.36	1.65	19.24
ShareGPT4V-7B [2]	**0.7144**	**0.8613**	0.30	1.65	16.31

Table 4. Effect of linguistic cues on saliency ranking performance.

Caption Variations	SA-SOR↑	SOR↑
Short captions (Human)	0.6856	0.8453
Short captions (LVLM)	0.6842	0.8488
w/o Adjectives/Adverbs	0.6877	0.8500
Shuffled Sentence Order	0.7028	0.8512
w/o Relational Expressions	0.6877	0.8517
Full Long-Form Descriptions	**0.7144**	**0.8613**

1.2.2. Effect of Linguistic Cues in Descriptions

Table 4 evaluates how caption source, length, and linguistic content affect saliency-ranking performance. Existing saliency-ranking datasets do not provide paired human-written long-form descriptions, preventing a direct comparison between human- and LVLM-generated descriptions of comparable richness. We therefore perform a controlled comparison using human-written MSCOCO captions [14] and short captions generated by LLaVA. MSCOCO captions contain an average of 10 tokens per caption. To generate the short captions, we use the same LVLM configuration as for the long-form descriptions, except that the maximum output length is limited to 15 tokens to approximately match the average length of the MSCOCO captions. The two settings yield closely comparable performance: 0.6856 versus 0.6842 in SA-SOR and 0.8453 versus 0.8488 in SOR. These results indicate that, under a matched-length setting, LVLM-generated captions can serve as a viable substitute for human-written captions in TSPNet.

Full long-form LVLM descriptions achieve the best SA-SOR and SOR scores, suggesting that richer object- and scene-level information provides useful auxiliary cues for saliency ranking. We further remove adjectives/adverbs and relational expressions, identified using spaCy [9], from the full descriptions. Both modifications reduce performance, suggesting that descriptive attributes and object–scene relations contain saliency-correlated information. Shuffling sentence order produces a smaller reduction, implying that the model relies more strongly on object-referential content than on global sentence ordering. Overall, these results suggest that the performance benefit primarily arises from the richness of the linguistic information, while the controlled short-caption comparison supports the practical use of LVLM-generated captions when human annotations are unavailable.

1.2.3. Effect of Hallucinations in Descriptions

This section details our controlled hallucinated-description generation and FaithScore-style verification protocols and analyzes TSPNet's performance under verified object, attribute, relation, and mixed hallucinations.

Hallucinated Description Generation. We adapt SugarCrepe's hard-negative generation strategy [10] to produce controlled candidate corruptions targeting objects, attributes, and relations. Each IRSR description is segmented into sentences, and, for each corruption setting, 30% of sentences eligible for the specified operation are randomly selected using a fixed seed. In the object and attribute settings, the selected sentences are divided approximately equally between addition and replacement; thus, each operation affects approximately 15% of eligible sentences. Following [10], we consider only replacement for relations because relation addition frequently produces linguistically implausible descriptions. The mixed setting draws from all five operations: object addition and replacement, attribute addition and replacement, and relation replacement. For each selected sentence, Qwen2.5-7B-Instruct identifies an explicit target concept and minimally adds or replaces it while preserving the remaining sentence structure and content. For example, an attribute replacement may change *'a man wearing a red shirt"* to *'a man wearing a blue shirt."* Importantly, the model receives only the sentence and has no access to the corresponding image. Consequently, the generated outputs are treated only as candidate corruptions: they may remain visually supported, fail to perform the intended operation, or introduce an actual hallucination. We therefore apply the independent visual-verification procedure described below before including them in the robustness evaluation.

FaithScore-Style Verification. We evaluate the original and candidate corrupted sentences separately using a FaithScore-style fact-decomposition and visual-verification pipeline [12]. Because the inputs are image-description sentences, Qwen2.5-7B [20] directly decomposes each sentence into atomic facts belonging to FaithScore's five categories: *entity*, *relation*, *color*, *count*, and *other attributes*. It performs only textual decomposition and does not assess

visual correctness. Qwen3-VL-8B [1] then receives the corresponding image and one atomic fact e_j at a time with the prompt: *"Fact: [FACT]. Is this fact visibly supported by the image? Answer exactly YES or NO."* We set $v_j = 1$ for `YES` and $v_j = 0$ for `NO`. The FaithScore, FS, is the proportion of facts judged visually supported:

$$\mathrm{FS}(D, I) = \frac{1}{N} \sum_{j=1}^{N} v_j, \tag{1}$$

where N is the number of atomic facts extracted from sentence D.

Because FaithScore measures faithfulness but does not define a corruption-acceptance rule, we retain a candidate when (i) its sentence-level FaithScore is lower than that of the original and (ii) it contains an unsupported fact belonging to the requested corruption category. For this matching, entity facts correspond to object corruptions; color, count, and other-attribute facts to attribute corruptions; and relation facts to relation corruptions. Candidates failing either criterion are reverted to their original sentences. We refer to the resulting descriptions as the *filtered description set* and the retained candidates as *filtered corruptions*. These automated judgments are used only for filtering and are further evaluated through the manual audit below.

Manual Hallucination Verification. To assess the reliability of the automatic filtering, a single annotator manually audited 200 randomly sampled filtered corruptions. For each sample, the annotator viewed the image, original sentence, corrupted sentence, and intended operation, and answered: (Q1) whether the original sentence was visually supported, (Q2) whether the corrupted sentence was visually supported, and (Q3) whether the intended addition or replacement was correctly performed. Each question allowed *Yes*, *No*, or *Uncertain*; uncertain responses were not counted as desirable outcomes.

Table 5 reports the results by corruption category and operation. Overall, 96.00% of the original sentences were judged image-supported, 80.50% of the filtered corruptions contained image-unsupported content, and 86.50% correctly performed the intended operation. Object corruptions were the most reliable: all were judged unsupported, and 96.77% correctly performed the intended operation. Attribute corruptions were also generally reliable while relation replacements were less reliable: only 54.41% were judged unsupported, and 72.06% correctly performed the intended operation.

In contrast, many relation replacements remained image-supported. Effective relation corruption requires identifying the connected entities and substituting a grammatically and semantically coherent relation while preserving the rest of the sentence. However, language models often favor plausible replacements that remain compatible with the entities and context, changing the wording without contradicting the main visual proposition. For example, replacing *'at"* with *'behind"* in *"A bus is awaiting its turn at a traffic light"* correctly changes the relation, but the bus may simultaneously be at and behind the traffic light. The replacement therefore does not necessarily create a clear visual contradiction. Object and attribute corruptions are less affected by this issue because many alternative objects or attributes can fit the same linguistic context while remaining absent from the image, enabling structurally coherent replacements that introduce clear visual contradictions.

Table 5. Manual audit of 200 FaithScore-filtered corruptions. N is the number of evaluated sentences. `add_obj` and `add_att` add an object and attribute, respectively; `replace_obj`, `replace_att`, and `replace_rel` replace an object, attribute, and relation, respectively. We report the desirable responses: *Yes* for Q1 and Q3 and *No* for Q2.

Category	**N**	**Q1 Yes (%)**	**Q2 No (%)**	**Q3 Yes (%)**
Object	62	100.00	100.00	96.77
Attribute	70	100.00	88.57	91.43
Relation	68	88.23	54.41	72.06
Operation	**N**	**Q1 Yes (%)**	**Q2 No (%)**	**Q3 Yes (%)**
`add_obj`	32	100.00	100.00	93.75
`replace_obj`	30	100.00	100.00	100.00
`add_att`	32	100.00	75.00	100.00
`replace_att`	38	100.00	100.00	84.21
`replace_rel`	68	88.23	54.41	72.06
Total	**200**	**96.00**	**80.50**	**86.50**

Result Analysis. Table 6 reports TSPNet's performance on the filtered set together with the filtering statistics. The hallucination experiments use a different checkpoint from the main IRSR ResNet-101 evaluation, explaining why their original-description baseline is 0.7216/0.8626 rather than 0.7144/0.8613. Within this experiment, the same checkpoint is used for all paired comparisons between original and corrupted descriptions. Candidate corruptions appear only in these statistics and are not used for performance evaluation. The full-test-set evaluation includes both descriptions containing at least one filtered corruption and descriptions left unchanged after filtering. To isolate the effect of the filtered corruptions, we also evaluate only the affected images, comparing their filtered-set descriptions with the corresponding original descriptions. Because the affected-image subset differs by corruption type, these paired results should be compared within each row rather than across rows.

On the full test set, all four filtered sets reduce SA-SOR relative to the original descriptions. SOR also decreases for the object, attribute, and mixed settings, while remaining unchanged after rounding for relations. Object corruptions

Table 6. Effect of filtered description corruptions on TSPNet. *Full Test Set* reports performance using the complete filtered set. *Affected Set* restricts evaluation to images containing at least one filtered corruption and compares their corrupted descriptions with the corresponding original descriptions. *Filtered* is the percentage of candidate corruptions retained after FaithScore-style filtering, and *Final* is the percentage of all test sentences represented by filtered corruptions. FS_{org} and FS_{corr} are the mean FaithScores of the original sentences and candidate corruptions, respectively, computed over all candidate pairs before filtering.

	Full Test Set		**Affected Set**				**Corruption Filtering Statistics**			
			Filtered Set		**Original Set**					
Type	**SA-SOR↑**	**SOR↑**	**SA-SOR↑**	**SOR↑**	**SA-SOR↑**	**SOR↑**	**Filtered (%)**	**Final (%)**	FS_{org}	FS_{corr}
Object	0.7102	0.8603	0.7353	0.8700	0.7363	0.8712	82.96	23.47	0.65	0.40
Attribute	0.7160	0.8609	0.7080	0.8558	0.7087	0.8577	62.04	17.69	0.65	0.51
Relation	0.7157	0.8626	0.7260	0.8676	0.7258	0.8679	82.24	20.11	0.64	0.53
Mixed	0.7126	0.8607	0.7164	0.8627	0.7212	0.8647	91.77	27.14	0.65	0.51
Original	0.7216	0.8626	–	–	–	–	–	–	–	–

produce the largest overall reduction (0.0114 SA-SOR and 0.0023 SOR), followed by mixed corruptions (0.0090 and 0.0019). These full-set changes are modest because filtered corruptions replace only 17.69–27.14% of the test sentences, leaving the remaining sentence content unchanged.

On the affected images, mixed corruptions produce the clearest reduction relative to the corresponding original descriptions (0.0048 SA-SOR and 0.0020 SOR). Object and attribute corruptions cause smaller reductions, whereas relation corruptions have a negligible effect. This relation result should be interpreted alongside the manual audit: only 54.41% of the filtered relation corruptions were judged image-unsupported, as many replacements remained compatible with the original visual proposition. It therefore reflects the limited strength of the generated relation perturbations rather than establishing insensitivity to genuine relational hallucinations. Overall, the results indicate that erroneous textual content can affect saliency ranking, particularly under object and mixed corruptions, while TSPNet remains relatively stable under the sparse corruption levels evaluated here.

1.3. CLIP-Based ROI-Text Matching Evaluation

We evaluate CLIP-based matching on 301 ROIs randomly sampled from the IRSR test set using a fixed seed of 42. For each ROI, CLIP selects the highest-scoring sentence from the corresponding image description. The sentence is retained when its similarity is at least $\tau = 0.2$; otherwise, an empty sentence is used. $\tau = 0.2$ was selected on validation data and fixed before this audit. For each retained match, two authors were independently shown the ROI, the full image, and the sentence selected by CLIP. They judged whether the sentence correctly and specifically referred to the ROI by selecting one of three options: *Correct*, *Incorrect*, or *Uncertain*. Table 7 summarizes the results.

Coverage measures how often CLIP produced a match above the threshold: 300 of 301 ROIs received a retained match, yielding 99.67% coverage. Because both annotators evaluated all 300 retained matches, the experiment produced 600 individual judgments. Of these, 469 were labeled *Correct*, giving an individual-judgment correctness rate of 78.17%. Exact three-class agreement measures how often both annotators selected the same label; they agreed on 255 of 300 pairs (85.00%). Cohen's κ was 0.57, indicating moderate agreement beyond chance according to the Landis–Koch interpretation scale [13]. For pair-level analysis, *Incorrect* and *Uncertain* are combined into a single *Non-correct* class. Under this binary grouping, the annotators agreed on 259 of 300 pairs (86.33%). This value is slightly higher than the exact three-class agreement because a pair labeled *Incorrect* by one annotator and *Uncertain* by the other is still considered a binary agreement. Among the 259 agreed pairs, both annotators labeled 214 as Correct. Thus, 82.63% of agreed pairs were jointly judged correct.

Table 7. Human evaluation of CLIP-based ROI-sentence matching.

Per-Annotator Results				
Annotator	**Correct**	**Incorrect**	**Uncertain**	**Precision**
Annotator 1	243	49	8	81.00%
Annotator 2	226	73	1	75.33%
All judgments	469	122	9	78.17%

Aggregate Results		
Measure	**Calculation**	**Result**
Coverage	300/301	99.67%
Exact three-class agreement	255/300	85.00%
Exact three-class disagreement	45/300	15.00%
Cohen's κ	–	0.57
Binary agreement	259/300	86.33%
Consensus pair-level precision	214/259	82.63%
Verified-correct ROI yield	214/301	71.10%

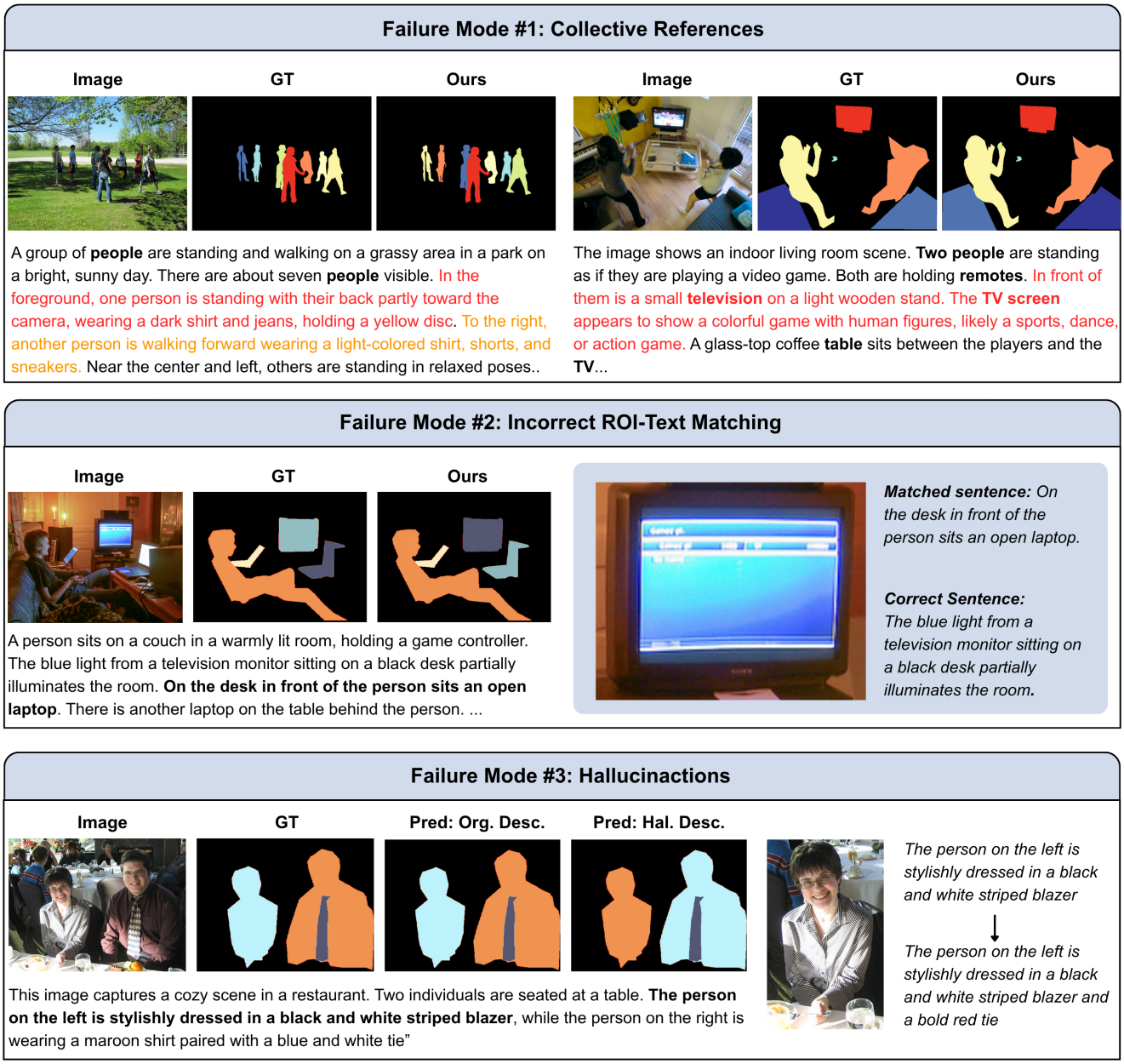


Figure 1. Representative failure modes and robustness examples of TSPNet. **Failure Mode #1:** In the left example, collective referring expressions and limited instance-specific visual separability result in incorrect rankings among people. In the right example, TSPNet recovers the correct saliency order despite collective language, suggesting that stronger visual contrast can compensate for limited instance-specific textual cues. **Failure Mode #2:** A sentence referring to the laptop is incorrectly CLIP-matched to the television ROI. In the corresponding prediction, the television is ranked below the laptop. **Failure Mode #3:** A hallucinated "bold red tie" causes the two people to be ranked incorrectly.

Finally, verified-correct ROI yield measures the proportion of all evaluated ROIs that received a retained match judged correct by both annotators; 214 of 301 ROIs met this criterion, corresponding to 71.10%.

1.4. Ablation of the MKTA Kernel Gating Factor

In this ablation, we analyze the sensitivity of the text-adaptation weight γ_t used to construct the MKTA kernels. Each text-adaptive kernel is defined as $W_t = w_t + \gamma_t \delta_t(s_j^{\text{enc}})$, where w_t is a learned base kernel, $\delta_t(s_j^{\text{enc}})$ is the text-conditioned residual generated from the matched

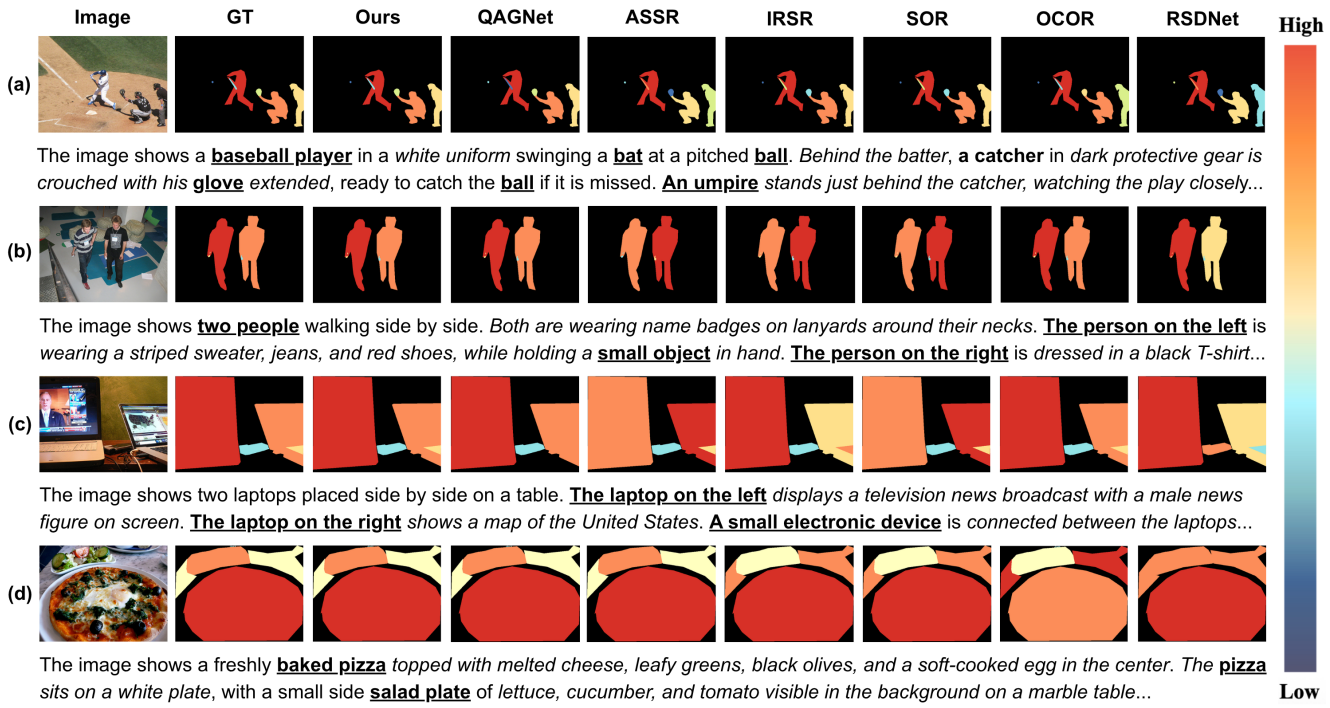


Figure 2. Additional qualitative comparisons on samples from the IRSR dataset.

Table 8. Ablation of the MKTA kernel formulation. Text only removes the base kernel; base only disables the text-conditioned residual; and the remaining variants combine both components using either a fixed or learned gate.

MKTA kernel formulation	SA-SOR	SOR
Text only ($w_t = 0, \gamma_t = 1$)	0.7035	0.8498
Base only ($\gamma_t = 0$)	0.7098	0.8563
Base + text ($\gamma_t = 1$)	0.7102	0.8556
Base + text (γ_t learned)	**0.7144**	**0.8613**

sentence, and γ_t is a scalar parameter defined separately for each kernel t that controls the residual's contribution.

Table 8 compares four kernel formulations. The text-only setting removes w_t and fixes $\gamma_t = 1$, constructing the kernel solely from the text-conditioned residual. Without a learned base kernel, this formulation may be less stable when the matched sentence is uninformative or empty. The base-only setting fixes $\gamma_t = 0$, reducing MKTA to learned depthwise kernels applied to the visual features. The two combined settings retain both components, with the per-kernel scaling parameter γ_t either fixed to 1 or learned during training. The text-only formulation performs worst, indicating the importance of retaining a stable base kernel. Learning γ_t produces the best results, suggesting that a learned weighting of the text-conditioned residual is more effective than a fixed unit weight.

1.5. Failure Cases and Boundary Conditions

Figure 1 illustrates three failure modes of TSPNet: collective references, CLIP-matching errors, and hallucinated descriptions. All predictions shown are produced using the same model checkpoint.

1.5.1. Failure Mode #1: Collective References

A key limitation arises when descriptions refer collectively to multiple same-category instances. Such a sentence may be validly matched to several ROIs, giving them similar textual evidence and limiting instance-level discrimination. When visual evidence is also ambiguous, the model may not reliably resolve their relative saliency. In the first example, the background people are described only as a group, providing no object-specific guidance and resulting in an incorrect ranking. In the second, despite partially collective language, the model recovers the correct ordering, suggesting that the visual stream can compensate when instances are sufficiently distinguishable. These examples suggest that TSPNet is vulnerable when language lacks instance-specific cues and visual evidence provides limited separability.

1.5.2. Failure Mode #2: Incorrect ROI-Text Matching

Incorrect CLIP-based ROI-text matching may provide an ROI with unreliable object-level textual evidence. Unlike collective-reference cases, where shared textual evidence may validly apply to multiple instances but lacks discriminative detail, this failure occurs when a sentence describ-

ing one object is incorrectly matched to a different ROI. In Fig. 1, a sentence describing the laptop is incorrectly matched to the television ROI. In the corresponding prediction, the television is ranked below the laptop.

1.5.3. Failure Mode #3: Hallucinated Descriptions

In Fig. 1, the same image is evaluated using an accurate original description and a corrupted description that hallucinates a "bold red tie" on the person on the left. The original description produces the correct ordering, whereas the corrupted description produces an incorrect ordering. This controlled example shows that hallucinated attributes can alter an individual prediction, although it does not indicate that textual evidence generally overrides the visual stream.

1.6. Additional Qualitative Comparison

Fig. 2 presents additional qualitative comparisons between TSPNet and the baseline models on representative samples from the IRSR dataset. These examples illustrate that TSPNet can produce more consistent saliency rankings and better preserve object-level distinctions across diverse scenes.

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